

KEVIN MITRE

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Education

City University of New York (CUNY) | Lehman College

Expected Graduation: Fall 2027

Bachelor of Science in Computer Science

New York, NY

- **Relevant Courses:** Data Structures and Algorithms, Object Oriented Programming, Database Systems, Linear Algebra

Experience

NYC Department of Social Services/Finance (DSS)

Sep 2025 – Dec 2025

Software Engineer Intern

Manhattan, NY

- Modernized the User Interface of the BARNS (**internal check monitoring system**), enabling deployment **two weeks ahead of schedule** by implementing **Bootstrap's responsive grid system**.
- Reviewed and refactored sections of the **FAST help-desk web application**, eliminating redundant code and simplifying logic to improve **code maintainability**, **readability**, and reduce **potential bugs**.
- Validated financial data using **SQL Server Management Studio (SSMS)**, identifying duplicate or inconsistent records to maintain accurate reporting.
- Presented technical updates to **business partners** via Microsoft Teams, translating development changes into clear project outcomes

Projects

Web Crawler | *Source Code*

C++ | Gumbo Parser | CurlCpp

- Collaborated on a **C++ web crawler project** to automatically scan websites and extract links using **curlcpp** and the **Gumbo HTML parser**, enabling automated **URL discovery and indexing** for further analysis.
- Implemented a **queue-based URL frontier** with **duplicate URL filtering** and concurrent crawling to efficiently traverse websites and avoid repeated page visits.
- Added **robots.txt compliance and link filtering** to ensure ethical crawling practices and prevent accessing restricted pages while improving crawler stability.

Snake Game | *Source Code*

Python | PyGame | PyTest

- Built a snake game using **Python** with **Pygame** and **Turtle**, implementing real-time movement, collision detection, and score tracking.
- Designed a game loop to process user input, update game state, and render graphics dynamically using event-driven programming.
- Wrote unit tests using **Pytest** to validate game logic, improving code reliability and ensuring correct behavior of core functions.

Blockchain Lottery | *Source Code*

Solidity | TypeScript | JavaScript

- Partnered with a group at **ETHGlobal NYC Hackathon** to develop a **decentralized lottery system** on **Ethereum**, enabling **provably fair and transparent draws** and automatic prize distribution.
- Apply **Chainlink Automation**, a decentralized oracle that provides trusted external data to smart contracts, to create a fully autonomous system that draws winners, distributes prizes, and starts new lottery rounds automatically.
- Engineered the contract logic to enforce participation thresholds, manage prize pools, and maintain a verifiable history of lottery outcomes and winners.

Skills

Languages: Python, MySQL, C/C++, C#

Developer Tools: .NET/ASP.NET Core, VS Studio, CLion, Pycharm, Git

Libraries/Frameworks: PyTest, BootStrap, Blazor